

CMOS1 HW2

ANALOGUE CMOS DESIGN 1

EE4520

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1 Table of results

Item	Design Item	Unit	Spec.	Achieved
1	Design Number (given to you)	–	80	80
2	SNR	[dB]	68.53	68.617
3	A_{settle}	[dB]	54	54
4	T_{settle}	[μ s]	0.56	0.559032
5	BW_{cl} from open-loop AC simulations	[MHz]	–	1.475
6	BW_{cl} from closed-loop AC simulations	[MHz]	–	1.452
7	Time-constant τ_{cl} (closed-loop AC)	[ns]	–	109.61
8	T_{40dB}	[μ s]	–	0.308274
9	$T_{48.69dB}$	[μ s]	–	0.441305
10	Time-constant τ_{cl} (transient)	[ns]	–	133.031
11	Power dissipation	[μ W]	Lowest possible	22.1006
12	$I(V_{dd})$	[μ A]	Lowest possible	12.2781
13	Total integrated noise	[μ V _{rms}]	–	316.86
14	Input step voltage	[mV]	$\frac{1.2}{8} = 150$	150
15	Output step voltage	[V]	1.2	1.2
16	C_{in}	[pF]	–	0.8218
17	C_{fb}	[pF]	$C_{in}/8$	0.1027
18	C_{load}	[pF]	C_{in}	0.8218
19	C_{CM}	[pF]	C_{fb}	0.1027
20	R_{big}	[G Ω]	Big	1000
21	C_{big}	[F]	Big	1
22	Bonus on FoM _{dB}	[dB]	0.08	0.08
23	$FoM_{lin} = \frac{2\pi \cdot Power \cdot T_{8.6dB}}{SNR_{lin}^2}$	[J]	$\leq 3.98\text{e-}18$	2.54e-18
24	$FoM_{dB} = -10 \log_{10}(FoM_{lin})$	[dB]	≥ 174.0	175.95

Table 1: Deliverable 1: Table of Results

2 Schematic with all node voltages

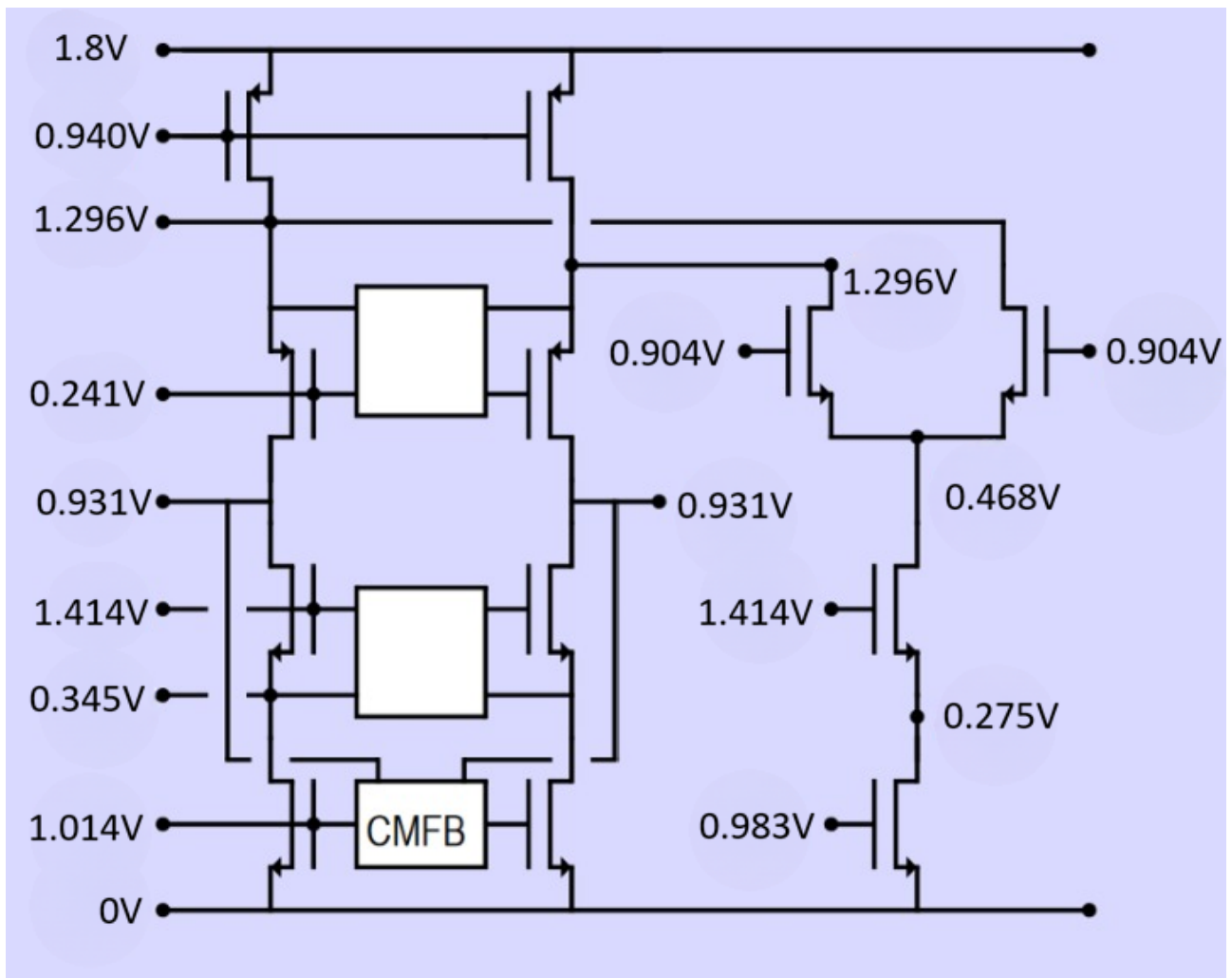


Figure 1: All node voltages

3 Schematic with all branch currents

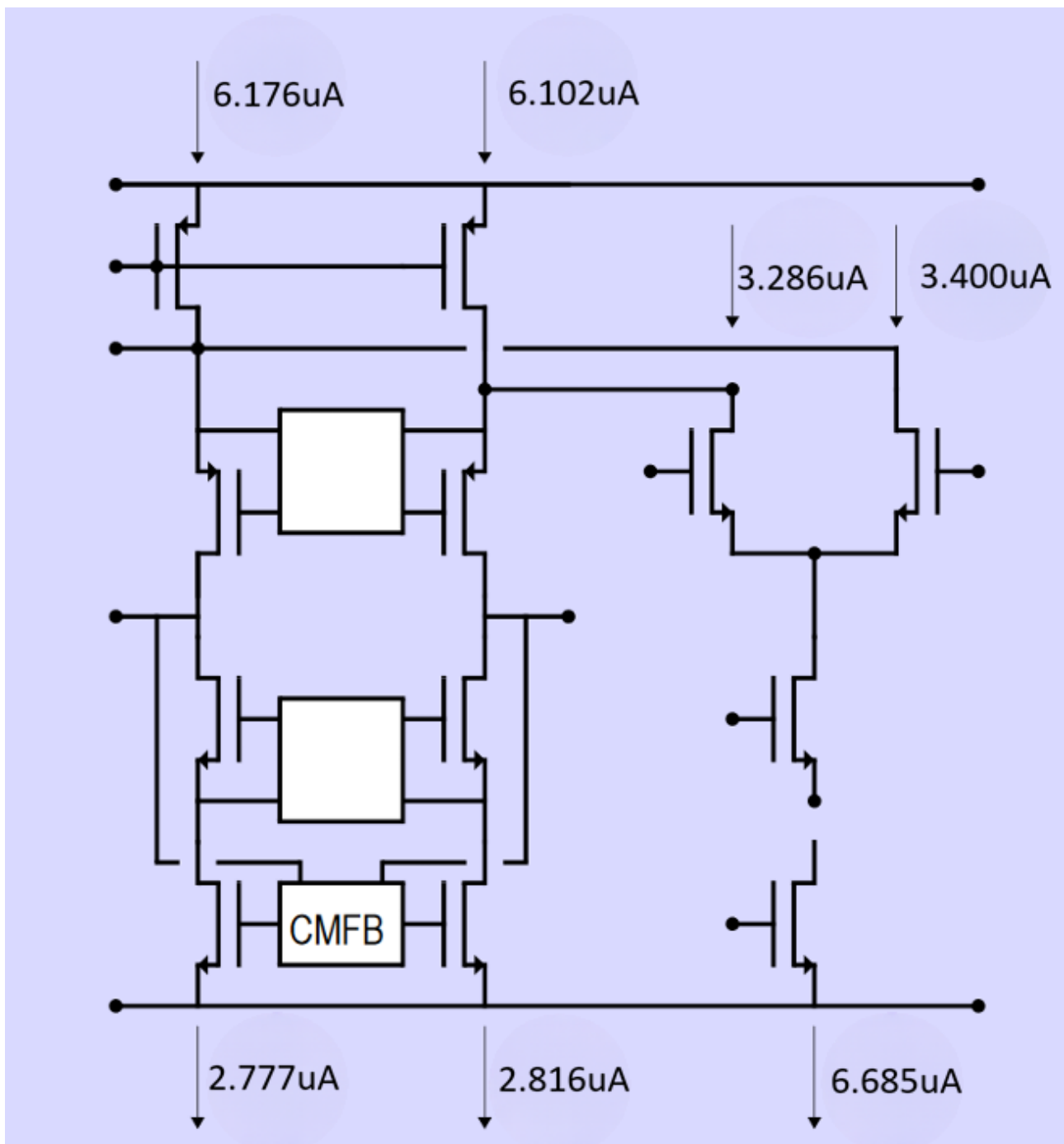


Figure 2: All branch currents

4 Bode-diagram of A and Aβ with Gain (dB) and Phase

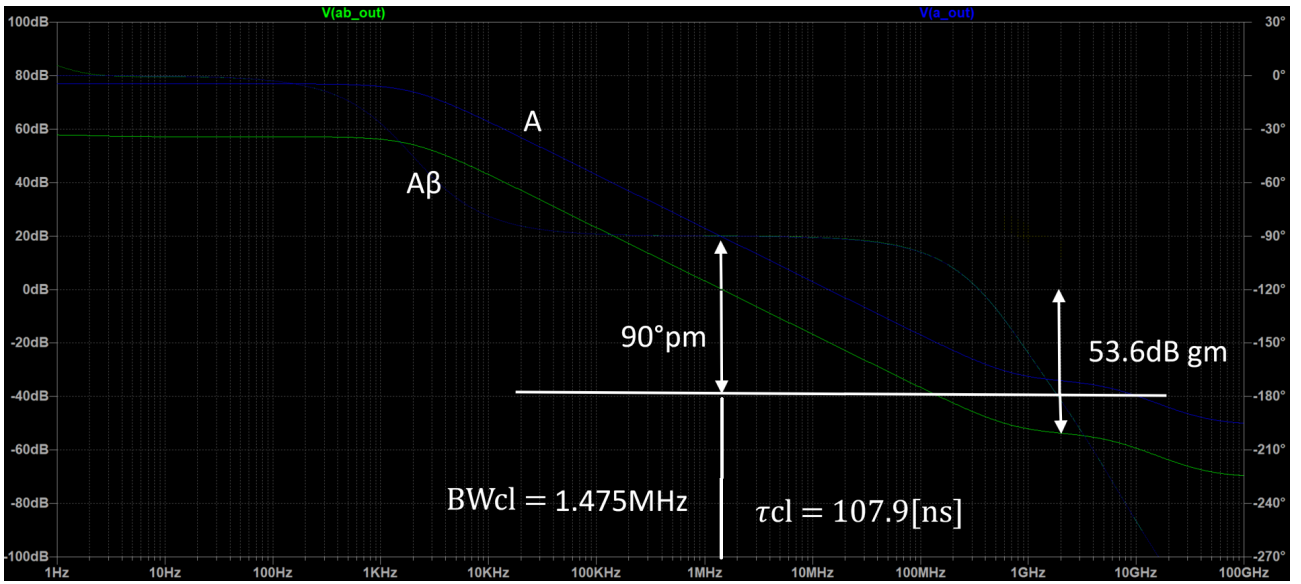


Figure 3: Bode-diagram of A and Aβ Unloaded

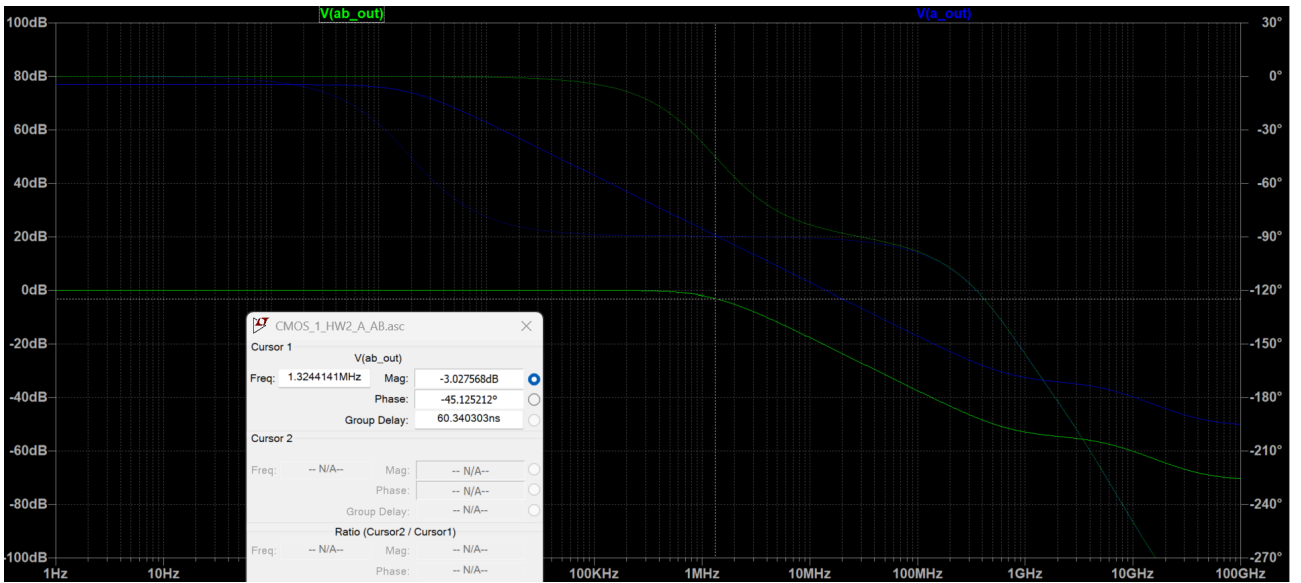


Figure 4: Bode-diagram of A and Aβ loaded

For the unloaded case shown in Figure 3, the phase margin is 90° and the gain margin is 53.6 dB. The unity-gain frequency of $A\beta$, which corresponds to the closed-loop bandwidth, is 1.475 MHz. The associated closed-loop time constant is therefore

$$\tau = \frac{1}{2\pi \cdot 1.475 \times 10^6} = 107.9[ns]$$

The constant separation between the gain plots of A and $A\beta$ is equal to $1/\beta$. In the loaded schematic shown in Figure 4, the open-loop gain A remains unchanged. However, $A\beta$ is constant at 0 dB up to the roll-off frequency, which is slightly lower than 1.475 MHz (the expected unity-gain frequency). This shift is caused by the parasitic capacitance introduced by the load amplifier. At low frequencies, $A\beta$ remains at 0 dB because the load amplifier in the feedback path acts like a buffer.

5 Bode-diagram of closed-loop Gain with Gain (dB) and Phase

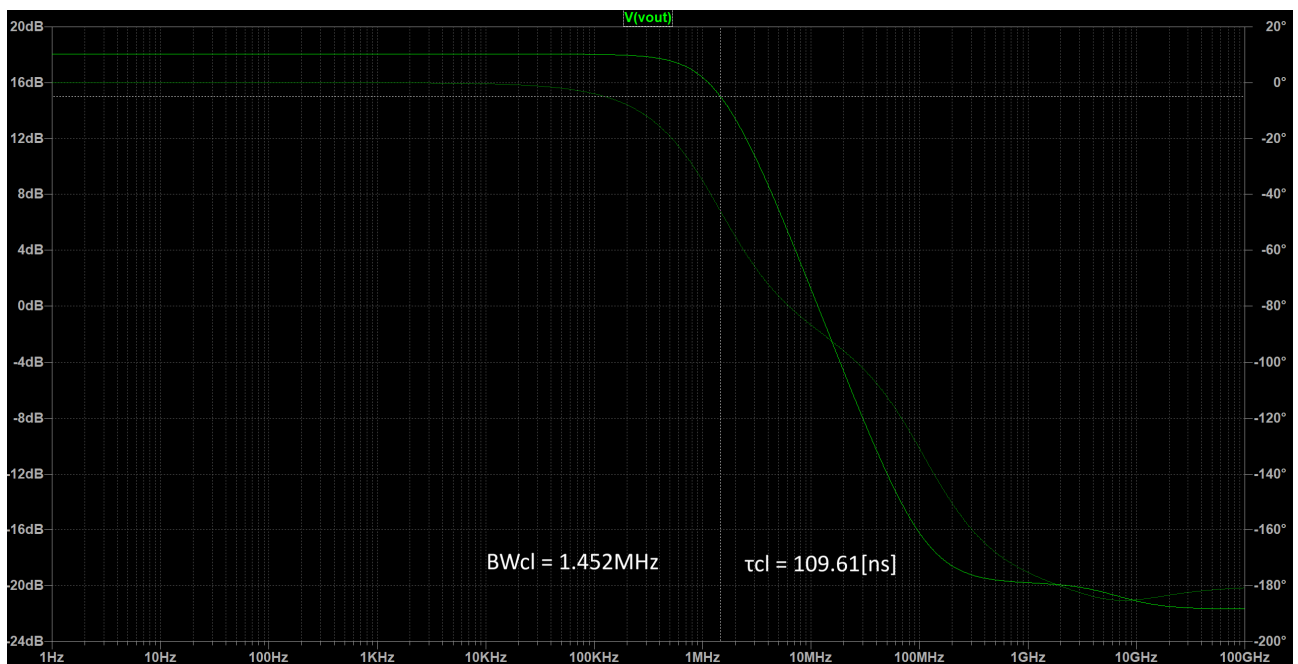


Figure 5: Bode-diagram of closed-loop Gain and phase

For the closed-loop gain, the maximum gain is approximately 18 dB, which corresponds to the intended gain of $8\times$. The cut-off frequency is almost identical to the value predicted from the open-loop simulation.

6 Settling accuracy versus time

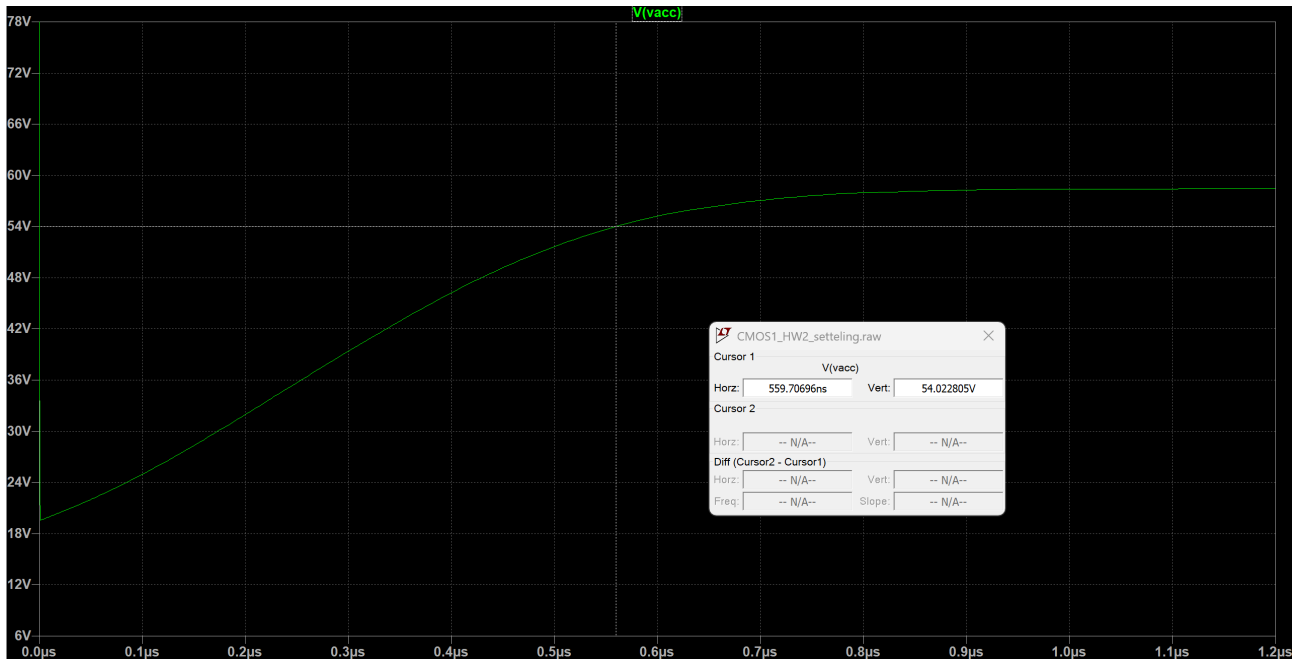


Figure 6:

$$\text{Accuracy}_{\text{dB}} = 20 \log_{10} \left(\frac{1.2 \text{ V}}{|V(\text{virtualground})| + 1 \text{ n}} \right)$$

Vs time

7 Settling accuracy versus time with estimated time-constant tcl



Figure 7: settling curve with times indicated

$$T_{\text{settle}} = 0.559\mu\text{s} < 0.56\mu\text{s}$$

$$T_{48.69\text{dB}} = 441.305 \text{ ns}$$

$$T_{40dB} = 308.274 \text{ ns}$$

$$\tau_{cl} = 133.031 \text{ ns}$$

The extracted closed-loop time constant τ_{cl} is higher than the value predicted by the AC simulation. This difference arises because the AC simulation is a small-signal linear analysis and does not account for the nonlinear effects present in the transient simulation.

8 Plot of output noise power density versus frequency

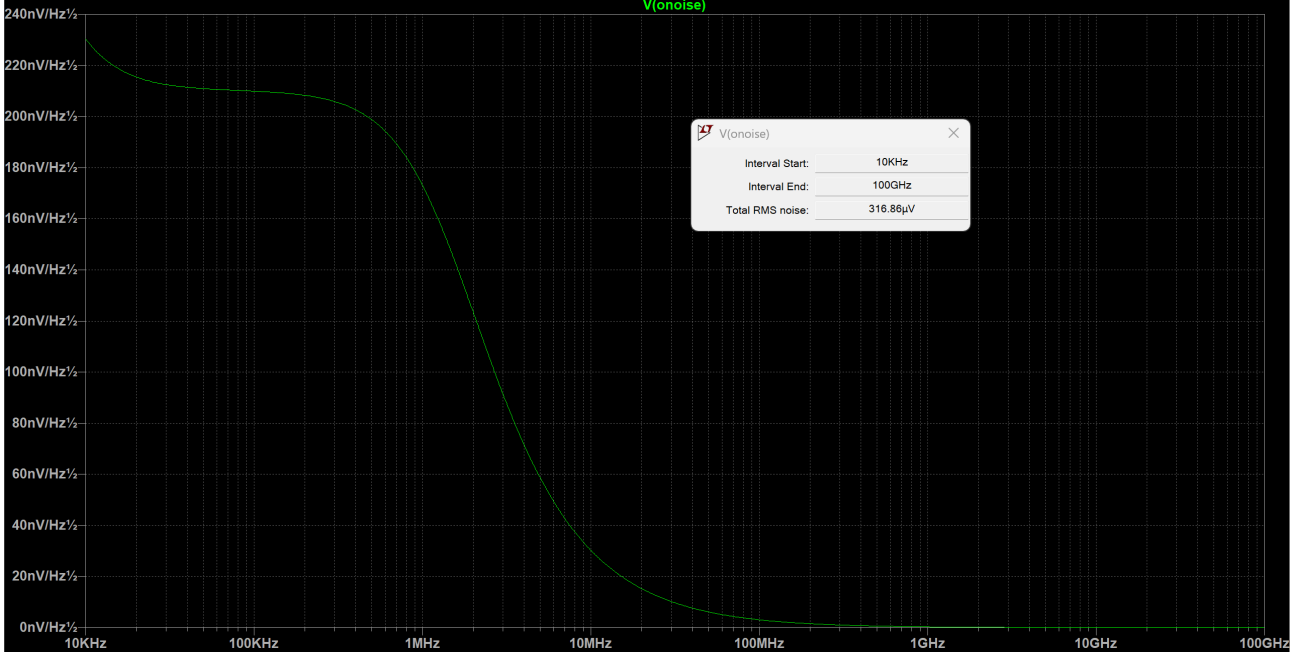


Figure 8: V_{Onoise} Vs frequency and integrated noise shown

9 Design Methodology

The first step in the design is to identify the available control variables. In this assignment we can modify M_n and M_p (NMOS/PMOS size ratios), r_n and r_p (ratios between I_{bmain} and the NMOS/PMOS bias currents), I_{bmain} , C_{in} (or C_{fb} and directly connected capacitances), $ScaleA$ (global transistor scaling factor), and in_ratio (input pair size ratio relative to the other transistors). Some of these variables are redundant. We fix M_n and M_p since they affect both the bias circuit and multiple devices in the amplifier, making them unnecessarily complex tuning parameters; full control is preserved through $ScaleA$ and r_n . We also fix r_p because only the relative ratio between r_n and r_p matters, while I_{bmain} remains free. The effective design variables are therefore I_{bmain} , r_n , C_{in} , in_ratio , and $ScaleA$.

Increasing I_{bmain} increases the overdrive voltage V_{GT} of all devices and increases the transconductance g_m . Since

$$\tau = \frac{1}{\beta\omega_u} = \frac{C_{tot}}{\beta g_{m,in}},$$

a larger $g_{m,in}$ improves settling speed. However, the open-loop gain

$$A_0 \approx g_{m,in} r_{out} = g_{m,in} g_{m,cas} r_{0cas1} r_{0cas2}$$

decreases with current because $g_m \propto \sqrt{I_D}$ while $r_o \propto \frac{1}{I_D}$, leading to $A_0 \propto \frac{1}{I_D}$. Therefore, increasing I_{bmain} reduces DC gain and limits steady-state accuracy since

$$\frac{A_0}{1 + A_0\beta} < \frac{1}{\beta}.$$

The optimal I_{bmain} is thus the largest value that still guarantees the required settling accuracy A_{settle} with sufficient margin, so that the steady state accuracy is above A_{settle} with some margin to not get into the slow part of the exponential $e^{-t/\tau}$ and have a hard time achieving T_{settle} .

The parameter in_ratio allows independent control of the input pair V_{GT} . Increasing in_ratio increases $g_{m,in}$ without proportionally increasing power, improving settling speed by reducing τ , and improving noise by decreasing the input pair V_{GT} , which reduces the noise excess factor. However, the effect on settling is more pronounced than on SNR, because beyond a certain point the noise excess factor is already sufficiently small. Therefore, in_ratio is increased (swept) until no significant improvement in settling is observed.

Finally, $ScaleA$ directly controls total power consumption. Reducing $ScaleA$ lowers power but modifies parasitic capacitances and internal pole locations. Although bandwidth can be recovered by adjusting C_{in} to maintain approximately constant C_{tot} , the noise spectrum exhibits an additional pole-zero behavior that is not solely determined by C_{tot} as seen in the left side of Figure 8, the noise curve isn't flat before the main pole set by C_{tot} . This pole-zero pair pulls up the noise graph when $ScaleA$ is decreased, thus increasing the integrated noise even if we maintain the same total C_{tot} . After optimizing I_{bmain} and in_ratio , the remaining tradeoff is between power (via $ScaleA$) and noise/settling (via C_{in}). The next design step is therefore to reduce $ScaleA$ to the minimum value that still satisfies both the settling and noise specifications. There remains one parameter to optimize: r_n . This parameter exhibits an optimum with respect to settling speed when the two output cascode transistors have approximately the same V_{DS} . When one of the cascode devices approaches the triode region, its output resistance r_o decreases. Since the effective cascode output resistance is proportional to $g_m r_{o1} r_{o2}$, a reduction in either r_o lowers the total output resistance. This leads to a decrease in open-loop gain and consequently slower settling. The maximum cascode resistance is therefore achieved when both devices are equally far from the triode region, i.e., when they share the available voltage headroom in a balanced manner. Finally, to achieve the best FoM, the previous steps should be iterated, since each parameter influences the optimal values of the others. For example, a significant change in $ScaleA$ shifts the optimal I_{bmain} , and similarly affects the optimal r_n and in_ratio . Therefore, the optimization must be performed iteratively until convergence to a consistent set of parameters that jointly satisfies the specifications and minimizes the power consumption.

Design Parameter	Final Value
M_n	11
M_p	65
r_n ($I_{b,ncas} = r_n I_{bmain}$)	2.73
r_p ($I_{b,pcas} = r_p I_{bmain}$)	6.3
I_{bmain}	1530 μA
in_ratio ($M_{n3,4} = in_ratio * ScaleA$)	4600
$ScaleA$	0.01302
C_{in}	0.8218

Table 2: Final optimized design parameters.